

Notes: Lileeva and Trefler (QJE, 2010)

Improved Access to Foreign Markets Raises Plant-Level Productivity... for Some Plants

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1 Big picture

- **Question.** Does improved access to foreign markets make domestic firms more productive, and if so, which firms benefit the most?
- **Setting.** The paper studies Canadian manufacturing plants around the Canada-U.S. Free Trade Agreement (FTA). The key policy shock is the elimination of U.S. tariffs on Canadian goods, together with lower Canadian tariffs on U.S. intermediate inputs.
- **Core challenge.** A raw correlation between exporting and productivity is not enough. More productive plants may self-select into exporting, and different plants may gain different amounts from exporting.
- **Main conceptual contribution.** The paper combines a simple model of exporting and investing in productivity with an instrumental-variables design that explicitly allows for heterogeneous treatment effects.
- **Model insight.** Exporting and investing in productivity are complements. Better foreign market access raises the market size over which productivity improvements pay off, so some firms respond by both exporting and investing.
- **Why heterogeneity matters.** The paper allows two dimensions of heterogeneity:
 - initial productivity,
 - and the productivity gain from investing.

This generates a strong prediction: among firms that newly start exporting, the productivity gains from trade should be largest for the initially less productive firms.

- **Main empirical design.** The authors build plant-specific tariff cuts using the exact HS6 products produced by each plant and use these tariff cuts as instruments for export growth.
- **Main result for 1984 nonexporters.** Canadian plants that were induced by tariff cuts to start exporting or export more increased labor productivity, adopted advanced manufacturing technologies, and engaged in more product innovation.
- **Main heterogeneity result.** These gains are concentrated among plants that were initially smaller and less productive. For the most productive plants, the estimated effect is close to zero.
- **Main mechanism result.** The same low-productivity plants that show labor-productivity gains are also the ones that adopt manufacturing information systems (MIS), inspection and communications technologies, and product innovation.
- **Main result for old exporters.** Plants that were already exporters before the FTA also benefit from improved market access, but unlike new exporters their treatment effect does *not* decline with initial productivity.
- **Main aggregate takeaway.** The paper argues that the FTA raised Canadian manufacturing productivity not only through reallocation across plants, but also through within-plant productivity improvements.
- **Economic takeaway.** Trade liberalization can raise plant productivity, but not uniformly. It mainly helps the firms for which foreign market access makes investing in productivity worthwhile.

2 Data and measurement (key objects)

2.1 Why Canada and the FTA are a useful laboratory

- FTA negotiations began in 1985, implementation started on January 1, 1989, and by 1996 the tariff on each tariff-line item was down to less than one-fifth of its 1988 level.
- Canadian manufacturing exports to the United States changed little during the negotiation period and the deep 1989–1991 recession, but then rose spectacularly after 1991.
- The paper does not study short-run export dynamics. Instead it studies the longer-run response between pre-FTA and post-FTA outcomes, using plant-level export data for 1984 and 1996.

Interpretation. The Canada-U.S. FTA gives a large, policy-driven improvement in foreign market access. Because the tariff schedule is external to any one plant, it provides a credible source of variation for exporting.

2.2 Plant-specific tariff cuts

For each plant, the authors observe every HS6 product it produces. They then:

1. start with U.S. statutory tariffs against Canadian imports at the HS8 level,
2. compute the change between 1988 and 1996,
3. aggregate to HS6 using 1996 U.S. import weights from Canada,
4. and average these product-level tariff changes across the products made by the plant.

This gives a plant-specific U.S. tariff cut, denoted $\Delta\tau$.

The tariff measure is then converted into mutually orthogonal binary instruments:

$$\Delta\tau_q, \quad q = 1, 2, 3, 4, \tag{1}$$

where the quartiles are based on the distribution of plant-level tariff cuts for 1984 nonexporters.

Interpretation. This is sharper than an industry-level tariff because two plants in the same industry can produce different mixes of goods and therefore experience different improvements in access to the U.S. market.

2.3 Main samples

The paper distinguishes two economically different groups.

- **1984 nonexporters.** These are 5,233 plants that did not export in 1984 and survived to 1996.
- **Old exporters.** These are 1,607 plants that exported in both 1984 and 1996.

For technology adoption and innovation, the authors use the 1993 Survey of Innovation and Advanced Technologies (SIAT), which covers 512 plants from the 1984-nonexporter sample.

Interpretation. These two samples match the model’s two key margins. Nonexporters face a joint decision about whether to export and invest; old exporters have already paid the fixed cost of exporting and only decide whether to invest more.

2.4 Key variables

Labor productivity growth. Let LP_t denote value added per worker in year t . The main dependent variable is

$$\Delta\phi \equiv \ln\left(\frac{LP_{1996}}{LP_{1988}}\right)/8. \quad (2)$$

Interpretation. This is the average annual log growth rate of labor productivity between 1988 and 1996.

Treatment for 1984 nonexporters. In the baseline new-exporter design,

$$T = \begin{cases} \ln(EXP_{1996}) & \text{for new exporters,} \\ 0 & \text{for nonexporters.} \end{cases} \quad (3)$$

The paper also considers a binary treatment $T \in \{0, 1\}$.

Treatment for old exporters. For plants that exported in both 1984 and 1996, the treatment is the average annual export growth:

$$T = \ln\left(\frac{EXP_{1996}}{EXP_{1984}}\right)/12. \quad (4)$$

Technology adoption. The SIAT survey records whether plants adopted advanced manufacturing technologies. The most important category is manufacturing information systems (MIS), which cover computer-based production management, scheduling, inventory, and machine loading systems. The paper also studies inspection and communications technologies.

Innovation. The SIAT survey records whether plants engaged in product or process innovation during 1989–1991, and separately whether they engaged in product innovation.

TFP-proxy checks. To address the limitation that labor productivity may not equal TFP, the paper also studies domestic sales growth, inventory reductions, intermediate-input purchases relative to shipments, and energy purchases relative to shipments.

2.5 Controls and bins

In the main specifications for 1984 nonexporters, the covariates are:

- log employment in 1984,
- log labor productivity in 1984,
- and annual labor productivity growth over 1984–1988.

To study heterogeneity, plants are grouped in bins based on within-industry productivity and employment quartiles. The paper’s baseline binning uses five groups formed from diagonal bands in the productivity-by-size matrix from Figure IV.

Interpretation. The bins are not arbitrary decoration. They are designed to map the model's prediction that responses should vary with initial productivity and size.

2.6 Descriptive facts

Table I reports several striking differences between new exporters and nonexporters before the FTA.

- New exporters were larger: log employment in 1984 is higher by 0.582.
- They were also more productive in 1984: log labor productivity is higher by 0.074.
- Their annual labor productivity growth was already higher before the FTA, by 0.030 over 1984–1988.
- By 1996 their export-to-sales ratio was higher by 0.197.
- They were also much more exposed to tariff cuts: the probability of receiving a positive tariff cut is higher by 26.5 percentage points, and the probability of receiving an above-median tariff cut is higher by 19.2 percentage points.

Table II shows that even in raw data, new exporters had faster labor-productivity growth than non-exporters, and that this difference declines monotonically across initial labor-productivity quartiles from 0.053 to 0.027 log points per year.

Interpretation. The raw facts already point to two things at once: first, tariff cuts matter for exporting; second, the benefit of exporting is not uniform across plants.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper embeds exporting and investing in productivity inside a simple monopolistic-competition model with CES demand.

A home firm with transformed productivity ϕ_0 chooses whether to export and whether to invest to raise productivity to ϕ_1 . When it exports, it pays a fixed cost F^E and faces foreign demand scaled by $\tau^{-\sigma} A^*$, where $\tau - 1$ is the ad valorem tariff.

Profits without productivity investment are

$$\pi_0(E) = \phi_0(A + E\tau^{-\sigma} A^*) - EF^E, \quad (5)$$

where $E \in \{0, 1\}$ indicates exporting.

Profits with productivity investment are

$$\pi_1(E) = \phi_1(A + E\tau^{-\sigma} A^*) - EF^E - F^I. \quad (6)$$

The gain from exporting and investing rather than doing neither is

$$\pi_1(1) - \pi_0(0) = \underbrace{[\phi_0\tau^{-\sigma} A^* - F^E]}_{\text{exporting only}} + \underbrace{[(\phi_1 - \phi_0)A - F^I]}_{\text{investing only}} + \underbrace{[(\phi_1 - \phi_0)\tau^{-\sigma} A^*]}_{\text{complementarity}}. \quad (7)$$

Interpretation. The last term is the core of the paper. If a firm both exports and becomes more productive, the productivity gain applies to foreign sales as well as domestic sales. That is why improved foreign market access can induce productivity-enhancing investment.

3.2 Selection line and negative selection

For firms that would otherwise choose between “export and invest” and “do neither,” indifference is given by

$$\phi_1 - \phi_0 = -\phi_0 \frac{\tau^{-\sigma} A^*}{A + \tau^{-\sigma} A^*} + \frac{F^I + F^E}{A + \tau^{-\sigma} A^*}. \quad (8)$$

Interpretation. This line slopes downward in initial productivity. Among firms that are just indifferent, the lower-productivity firms must expect larger gains from investing in order to find exporting worthwhile. This is the source of the paper’s “negative selection” prediction for new exporters.

3.3 Figure I and Figure II logic

The model predicts three types of firms after a fall in foreign tariffs:

- some firms start exporting and investing,
- some start exporting without investing,
- and some firms that already exported now invest more.

For firms that *start* exporting, the productivity effect should be decreasing in initial productivity. For firms that *already exported*, the productivity effect should be approximately homogeneous.

Interpretation. This distinction between new exporters and old exporters is not an afterthought. It is one of the cleanest implications of the model, and the empirical analysis is built to test it directly.

3.4 Standard IV specification

The paper first presents a textbook IV setup for 1984 nonexporters:

$$\Delta\phi = \beta T + \gamma X + \varepsilon, \quad (9)$$

$$T = \sum_{q=2}^4 \delta_q \Delta\tau_q + \gamma' X + \eta, \quad (10)$$

where X includes 1984 employment, 1984 labor productivity, and 1984–1988 productivity growth.

Interpretation. The coefficient β measures how labor productivity growth relates to the export growth induced by tariff cuts. The paper estimates this separately by bin to allow treatment effects to vary with initial conditions.

3.5 Why the paper moves beyond standard IV

Once treatment effects are heterogeneous, standard IV no longer has a simple constant-effect interpretation. With a binary instrument, the local average treatment effect (LATE) can be written

as

$$\text{LATE} = \beta + \frac{E[U \cdot \Delta T]}{E[\Delta T]}, \quad (11)$$

where U is the plant-specific heterogeneous treatment component.

Interpretation. IV puts more weight on plants whose export response is moved more by the tariff instrument. In this paper, those are exactly the firms near the export-and-invest margin. So the empirical object is naturally a weighted average treatment effect for the compliers induced to export by the FTA.

3.6 Angrist-Imbens heterogeneous-response specification

To restore a weighted-average interpretation with covariates and multivalued treatments, the paper follows Angrist and Imbens (1995). It:

- converts the tariff instrument into binary quartile dummies,
- converts the three covariates into quartile dummies,
- and interacts these covariate dummies with the tariff dummies.

The second stage is

$$\Delta\phi = \beta T + \sum_{k=1}^3 \sum_{q=2}^4 \gamma_{kq} X_{kq} + \theta_{SIC} + \varepsilon, \quad (12)$$

and the first stage is

$$T = \sum_{q=2}^4 \delta_q \Delta\tau_q + \sum_{k=1}^3 \sum_{q=2}^4 \gamma'_{kq} X_{kq} + \sum_{k=1}^3 \sum_{q=2}^4 \sum_{q'=2}^4 \lambda_{kqq'} X_{kq} \Delta\tau_{q'} + \theta_{SIC} + \eta. \quad (13)$$

Interpretation. This is the paper's preferred estimator because it respects the fact that tariff-induced export responses differ systematically across observable plant types.

3.7 Technology adoption and innovation regressions

For the SIAT outcomes, the paper reuses the same second-stage and first-stage logic, but replaces labor productivity growth with binary outcomes for:

- MIS adoption,
- inspection and communications adoption,
- any innovation,
- and product innovation.

Because the SIAT subsample is smaller, plants are grouped into a **low bin** (baseline bins 1–3) and a **high bin** (baseline bins 4–5).

Interpretation. The idea is simple and powerful: if the same plants that gain in productivity are also the ones that adopt technologies and innovate, then the productivity results are much more plausibly driven by active investment rather than by accounting artifacts.

3.8 Old exporters specification

For old exporters the treatment is export growth between 1984 and 1996,

$$T = \ln \left(\frac{EXP_{1996}}{EXP_{1984}} \right) / 12. \quad (14)$$

The empirical question is no longer whether tariff cuts induce a firm to start exporting. It is whether improved market access makes existing exporters expand exports and invest more.

Interpretation. The theory predicts no negative selection here because the fixed cost of exporting has already been paid. The decision is only whether to invest more in productivity.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the causal effect

The main instrument is the plant-specific reduction in U.S. tariffs caused by the FTA. The identifying logic is:

- tariff cuts are imposed by trade policy, not chosen by plants;
- plants differ in exposure because they produce different goods;
- these differences in exposure predict which plants are induced to export more.

Interpretation. This is a shift-share style source of variation, but at the plant level rather than the industry level. The product mix that each plant already had before the FTA determines how strongly it is hit by the policy.

4.2 Why reverse causality is weak

A plant's productivity growth cannot cause the United States to change its statutory tariff schedule against Canadian products. The timing and the detailed schedule of tariff reductions are external to the individual plant.

Interpretation. This is the main reason the tariff-cut instrument is credible. Export growth may be endogenous, but tariff reductions are not.

4.3 Why the first stage is strong

The tariff cuts strongly predict exporting among 1984 nonexporters.

- 85% of new exporters received a tariff cut, versus 56% of nonexporters.
- 64% of new exporters received an above-median tariff cut, versus 40% of nonexporters.

In Table III, the first-stage coefficients on the tariff-cut dummies are positive and statistically significant across the relevant bins.

Interpretation. If the instrument barely moved exporting, the whole empirical design would be weak. But the first-stage numbers are economically and statistically strong, especially in the lower bins where most of the action is.

4.4 Why heterogeneity is part of identification, not a nuisance

The paper argues that the treatment effect is not a universal constant. Instead, the instrument identifies the productivity gain for the firms whose exporting behavior is shifted by the tariff cuts.

Interpretation. This is not a flaw in the design. It is exactly the point of the paper. The economically relevant question is: *which firms respond to improved foreign market access by both exporting and investing?*

4.5 Controlling for preexisting differences

New exporters are already larger and more productive before the FTA. The paper deals with this in three ways:

- industry fixed effects,
- controls for initial size and productivity,
- and separate estimation by productivity-size bins.

Interpretation. The empirical results are not based on pretending exporters and nonexporters looked the same ex ante. They did not. The strategy is instead to compare like with like as closely as possible and then let the tariff instrument isolate the export margin.

4.6 Alternative channels from Canadian tariff cuts

The FTA also lowered Canadian tariffs on U.S. intermediate inputs. This could itself improve productivity by lowering the cost of investing in productivity. The paper therefore adds:

- Canadian tariff cuts on intermediate inputs, $\Delta\tau^{Input}$,
- Canadian tariff cuts on final goods, $\Delta\tau^{Output}$,
- and the growth in Canadian imports from the United States, $\Delta \ln M$,

as additional first-stage instruments and robustness controls.

Interpretation. This matters because the reduced-form impact of the FTA is not only “sell more abroad.” It is also “buy better inputs from abroad.” The results survive these extra controls.

4.7 Selection issues: plant deaths and temporary exporters

The sample keeps plants alive from 1984 to 1996. The paper notes two potential concerns:

- some plants may start exporting and then stop,
- and some plants may die.

The authors argue these issues do not overturn the results. In particular, plant deaths tend to be concentrated among nonexporters with poor productivity growth, so accounting for deaths would if anything make the estimated effects larger.

Interpretation. This is useful because it means the main estimates are unlikely to be mechanically inflated by survivorship.

4.8 The big limitation: labor productivity is not TFP

The authors are explicit that they do not observe capital stock, so they cannot directly estimate TFP. Their main productivity measure is value added per worker.

Interpretation. This is the most important weakness of the paper. The authors do not hide it. Instead they bring in extra outcomes to show that the productivity results line up with what one would expect if true efficiency improved.

4.9 What the paper can and cannot distinguish

The paper is very persuasive on whether improved foreign market access induced some firms to invest and become more productive. It is less precise on:

- the exact decomposition between product innovation, process change, and management change,
- the exact TFP magnitude because capital is unobserved,
- and high-frequency export dynamics because exports are only observed in 1984 and 1996.

Interpretation. The paper's strongest contribution is the causal link from market access to productivity-enhancing investment for compliers, not a complete structural decomposition of all mechanisms.

5 Tables and figures

5.1 Figure I and Figure II: the model's partition of firms

Read:

- Figure I maps the firm's choices in the space of initial productivity ϕ_0 and productivity gain from investing $\phi_1 - \phi_0$.
- Figure II shows what happens after a fall in foreign tariffs: some firms start exporting and investing, some start exporting without investing, and some old exporters invest more.
- The shaded regions in Figure II make the key heterogeneity result visual: for new exporters, the productivity gain is largest for the less productive firms; for old exporters, the gain is roughly constant.

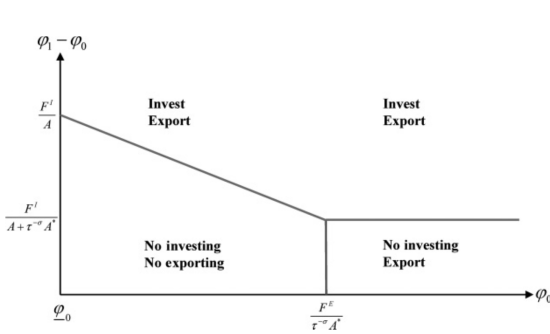


FIGURE I
The Optimal Choices of Exporting and Investing

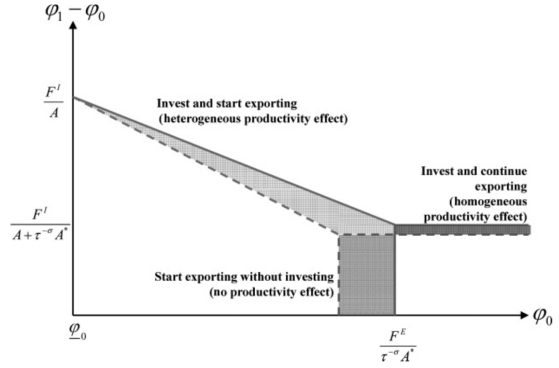


FIGURE II
Switching Behavior Induced by Improved Foreign Market Access

Economic message. These two figures are the theoretical backbone of the whole paper. They tell you in one picture why the paper keeps separating new exporters from old exporters and why the effect should be strongest among low-productivity starters.

5.2 Figure III and Table I: the FTA shock and the basic samples

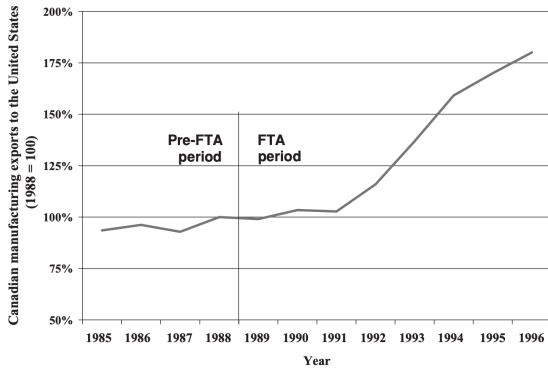


FIGURE III
Canadian Manufacturing Exports to the United States

TABLE I
AVERAGE PLANT CHARACTERISTICS

	New exporters less nonexporters		Old exporters less new exporters	
	(1)	(2)	(3)	(4)
Log employment, 1984	0.582	(21.52)	0.739	(22.12)
Log labor productivity, 1984	0.074	(4.72)	0.058	(2.98)
Annual labor productivity growth, 1988–1996	0.023	(8.83)	−0.001	(−0.42)
Annual labor productivity growth, 1984–1988	0.030	(6.91)	0.011	(2.02)
Exports/sales, 1996	0.197	(37.24)	0.148	(22.62)
Proportion of plants with $\Delta\tau > 0$	0.265	(23.00)	−0.030	(−2.08)
$\Delta\tau > \text{median}$	0.192	(18.80)	−0.069	(−5.59)
Canadian tariff cut on Final goods ($\Delta\tau^{\text{Output}}$)	0.020	(14.99)	0.002	(1.06)
Intermediate inputs ($\Delta\tau^{\text{Input}}$)	0.021	(22.97)	0.001	(0.46)

Notes: This table reports differences in means across groups of plants after controlling for industry fixed effects. There are three groups of plants: (1) plants that did not export in either 1984 or 1996 (nonexporters, $N = 3,114$); (2) plants that did not export in 1984, but exported in 1996 (new exporters, $N = 2,119$); and (3) plants that exported in both 1984 and 1996 (old exporters, $N = 1,607$). Each row reports the results of a regression which pools all three groups. The dependent variable is listed in the leftmost column. The independent variables are four-digit SIC fixed effects and three dummy variables indicating whether the plant is a nonexporter, new exporter, or old exporter. Column (1) reports the difference between the new exporter and nonexporter dummies. Column (3) reports the difference between the old exporter and new exporter dummies. t -statistics appear in parentheses.

Read:

- Figure III shows that Canadian manufacturing exports to the United States are flat through the negotiation period and the recession, then surge after 1991.
- Table I shows that new exporters are already different from nonexporters: they are larger, more productive, and faster-growing before the FTA.
- Table I also shows that the U.S. tariff cuts are highly correlated with export entry and that new exporters are more exposed to Canadian tariff cuts on U.S. intermediate inputs.

Economic message. Figure III establishes that a major export expansion happened during the FTA period. Table I then shows that the plants entering export markets were not random. That

is exactly why the paper needs an instrument rather than a simple exporter-versus-nonexporter comparison.

5.3 Table II and Figure IV: heterogeneity is visible before any econometrics

TABLE II
LABOR PRODUCTIVITY GROWTH, 1988–1996: NEW EXPORTERS LESS NONEXPORTERS

	Labor productivity quartiles in 1988			
	(1)	(2)	(3)	(4)
New exporters less nonexporters	0.053 (9.34)	0.036 (8.21)	0.030 (6.93)	0.027 (5.45)
Average number of employees per plant	53	74	95	93

Notes. The first row of this table reports the difference between new exporters and nonexporters in average annual labor productivity growth ($\Delta\varphi$). t -statistics are in parentheses. The first (last) column deals with plants in the lowest (highest) labor-productivity quartile. The sample consists of 1984 nonexporters.

		Labor productivity quartiles in 1988			
		1	2	3	4
Employment quartiles in 1988	1	0.061	0.052	0.050	0.041
	2	0.045	0.039	0.030	0.023
	3	0.057	0.033	0.021	0.013
	4	0.022	0.015	0.008	0.005

FIGURE IV
Labor Productivity Growth, 1988–1996: New Exporters Less Nonexporters by Productivity and Size

As in Table II, each cell is the difference between new exporters and non-exporters in average annual labor productivity growth ($\Delta\varphi$). Boldface indicates statistically significant differences at the 1% level. t -statistics appear in Online Appendix Table B.3. The sample consists of 1984 nonexporters.

Read:

- Table II shows that new exporters have faster labor-productivity growth than nonexporters in every initial productivity quartile, but the gap declines monotonically from 0.053 to 0.027 log points per year.
- Figure IV goes further by cross-classifying plants by both productivity and employment. The largest differences appear in the upper-left cells: small, less productive plants.
- The lower-right cells, corresponding to large and more productive plants, show much smaller and often insignificant differences.

Economic message. This is one of the most persuasive descriptive pieces of evidence in the paper. Before IV, before LATE, before any complex specification, the data already look like the model: new-exporter gains are strongest for the initially weaker plants.

5.4 Table III and Table IV: standard IV already shows economically large heterogeneous effects

TABLE III LABOR PRODUCTIVITY GROWTH, 1988–1996: STANDARD IV ESTIMATION																	
Bin	Treatment: $T = \ln(\text{EXP}_{\text{new}})$		Labor productivity		Labor prod. growth 1984–1988		Employment size		Tariff cut instruments						Alternative bias $T = \ln(\text{EXP}_{\text{non}})$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)
OLS dependent variable: Productivity growth																	
All	0.0022	(13.01)	-0.04	(-22.58)	-0.28	(-35.47)	0.01	(5.55)									
1	0.0042	(8.12)	-0.05	(-7.40)	-0.25	(-10.27)	0.01	(2.04)								0.0040	(9.94)
2	0.0041	(9.64)	-0.05	(-9.40)	-0.33	(-15.74)	0.00	(-0.50)								0.0036	(8.56)
3	0.0027	(6.67)	-0.05	(-10.44)	-0.32	(-18.03)	0.00	(1.39)								0.0023	(6.78)
4	0.0013	(3.48)	-0.04	(-8.45)	-0.35	(-14.45)	0.01	(3.36)								0.0018	(2.42)
5	0.0008	(2.79)	-0.03	(-7.19)	-0.30	(-9.16)	0.01	(3.64)								0.0002	(0.43)
Second-stage IV dependent variable: Productivity growth																	
All	0.010	(15.92)	-0.05	(-25.33)	-0.31	(-37.77)	-0.01	(-6.01)									
1	0.017	(9.87)	-0.05	(-7.88)	-0.24	(-10.06)	0.00	(-0.70)								0.016	(10.11)
2	0.015	(10.30)	-0.06	(-10.32)	-0.32	(-15.35)	-0.02	(-4.89)								0.014	(10.07)
3	0.012	(7.72)	-0.05	(-10.45)	-0.30	(-16.90)	-0.01	(-3.43)								0.010	(7.68)
4	0.008	(4.57)	-0.03	(-7.14)	-0.34	(-11.90)	0.00	(-0.19)								0.006	(4.00)
5	0.003	(2.44)	-0.03	(-7.25)	-0.20	(-9.28)	0.01	(2.79)								0.001	(0.85)
First-stage IV dependent variable: $T = \ln(\text{EXP}_{\text{new}})$																	
All	1.68	(20.48)	2.50	(4.03)	0.78	(5.03)	3.1	(12.04)	4.4	(18.82)	3.1	(13.01)					
1	0.13	(0.35)	-0.50	(-0.34)	0.50	(2.61)	1.8	(4.21)	3.1	(7.64)	2.9	(7.33)					
2	0.75	(1.80)	-0.70	(-0.46)	1.36	(5.99)	3.4	(6.10)	4.1	(8.53)	2.9	(5.92)					
3	-0.04	(-0.13)	-1.39	(-1.12)	1.63	(7.94)	3.4	(6.02)	4.3	(8.19)	3.2	(6.47)					
4	-0.45	(-1.14)	-0.97	(-0.23)	1.36	(6.12)	3.2	(3.27)	4.0	(6.54)	2.5	(4.65)					
5	0.33	(0.76)	1.75	(0.79)	0.64	(2.86)	3.2	(4.93)	4.6	(7.37)	2.4	(3.67)					

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Notes. The top panel of this table reports OLS estimates of the productivity growth equation (5). The dependent variable $\ln\varphi$ is the average annual log change in labor productivity over the period 1988–1996. The top row reports the treatment variable $T = \ln(\text{EXP}_{\text{new}})$ for new exporters and $T = 0$ for nonexporters. The middle panel reports the IV estimates of the productivity growth equation (6). The bottom panel reports estimates of the first-stage regression of equation (6) on the instruments. The first row of each panel reports the sample size N . The sample size N is 1,984. Columns (1)–(5) report the coefficient on the treatment T for the case where the bias is induced using the probit-based method. The sample variance of this regression is 0.0001. Columns (6)–(10) report the coefficient on the treatment T for the case where the bias is induced using the probit-based method. The sample variance of this regression is 0.0001. Columns (11)–(15) report the coefficient on the treatment T for the case where the bias is induced using the probit-based method. The sample variance of this regression is 0.0001. Columns (16)–(18) report the coefficient on the treatment T for the case where the bias is induced using the probit-based method. The sample variance of this regression is 0.0001. Columns are in parentheses.

TABLE IV											
SPECIFICATION TESTS AND COEFFICIENT MAGNITUDES											
Bin	Coefficient magnitudes			Hausman test	Over-id test		First stage F -tests				
	ΔT	$\beta \times \Delta T \times 8$	Emp. wgt.				3 tariffs		All variable		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
1	1.43	0.196	0.06	34.62	.00	3.56	.03	25.50	.00	18.28	.00
2	2.20	0.264	0.10	34.43	.00	1.53	.22	29.10	.00	32.68	.00
3	2.71	0.267	0.18	25.36	.00	0.05	.95	26.20	.00	43.88	.00
4	2.25	0.146	0.25	11.22	.02	1.40	.25	14.35	.00	22.88	.00
5	2.77	0.071	0.40	3.26	.52	2.47	.09	18.89	.00	18.46	.00
Total		0.153									

Notes. Column (1) is the estimated impact of the U.S. tariff cut on the treatment T , that is, on exporting. It is based on the estimates of equation (6) reported in the bottom panel of Table III. Column (2) is the impact of exporting on labor productivity for those plants that were induced to export as a result of the U.S. tariff cuts. The "Total" row reports the average of column (2), averaged using the column (3) employment weights. The table also provides specification tests for the IV specification that appears in the middle panel of Table III. Column (4) reports the Hausman χ^2 statistic for the difference between the OLS and IV estimates of four parameters (coefficients on T and three covariates). Column (6) reports the Basman (1960) χ^2 statistic for overidentification. Column (8) reports the F_N^H statistic for the joint null that all three tariff coefficients in the first stage equal 0. (Bin sample sizes N are given in the notes to Table III.) Column (10) reports the F_N^H statistics for the joint null that all six coefficients in the first stage equal 0. p -values appear in columns (5), (7), (9), and (11).

Read:

- Table III reports the simple IV estimates by bin. The treatment coefficient declines from 0.017 in bin 1 to 0.003 in bin 5.
- Table IV turns these coefficients into economically interpretable magnitudes by multiplying by the tariff-induced export change and the eight-year horizon.
- The employment-weighted average effect is 0.153 log points, meaning that improved access to the U.S. market raises labor productivity by about 15.3% for plants induced to export.
- The bin-level treatment effects are especially large in bins 1 to 3 and much smaller in bins 4 and 5.

Economic message. The standard IV results are already strong: the FTA-induced export response produces large within-plant productivity gains, but the gains are highly uneven. The more subtle Angrist-Imbens estimator refines the interpretation; it does not overturn the basic economic message.

5.5 Table V: the preferred Angrist-Imbens estimates

TABLE V
LABOR PRODUCTIVITY GROWTH 1988-1996, ANGRIST-IMBENS IV ESTIMATOR

Bin	IV					OLS			Alternative bins	
	β	t	ΔT	$\beta \times \Delta T \times 8$	Overidentification	β	t	R^2	β	t
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(1') (2')
A. Baseline: Five bins based on productivity and size; $T = 0, \ln(\text{EXP}_{1988})$										
1	0.012	(7.34)	1.58	0.147	(.22)	0.0041	(7.32)	.19	0.013	(9.65)
2	0.010	(7.77)	2.85	0.237	(.04)	0.0041	(8.29)	.20	0.012	(9.45)
3	0.009	(6.21)	3.46	0.241	(.15)	0.0031	(6.71)	.25	0.009	(8.18)
4	0.005	(2.54)	2.02	0.085	(.77)	0.0011	(2.76)	.20	0.002	(1.19)
5	0.002	(1.01)	1.51	0.022	(.29)	0.0007	(1.97)	.16	0.002	(1.65)
Total				0.107						
B. Five bins based on productivity and size; binary treatment ($T = 0, 1$)										
1	0.154	(7.34)	0.12	0.149	(.24)	0.0519	(6.99)	.18	0.183	(9.76)
2	0.140	(7.71)	0.21	0.234	(.04)	0.0534	(7.86)	.20	0.159	(9.43)
3	0.117	(6.06)	0.24	0.228	(.12)	0.0388	(6.08)	.24	0.129	(8.14)
4	0.084	(2.93)	0.14	0.091	(.88)	0.0144	(2.46)	.20	0.021	(1.21)
5	0.021	(0.88)	0.11	0.018	(.27)	0.0081	(1.59)	.16	0.032	(1.47)
Total				0.105						

TABLE V
CONTINUED

Bin	IV					OLS			Alternative bins	
	β	t	ΔT	$\beta \times \Delta T \times 8$	Overidentification	β	t	R^2	β	t
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(1') (2')
C. Four bins based on productivity; $T = 0, \ln(\text{EXP}_{1988})$										
1	0.010	(8.27)	2.88	0.240	(.02)	0.0036	(7.86)	.20	0.011	(10.48)
2	0.009	(7.66)	3.34	0.251	(.01)	0.0028	(8.10)	.13	0.008	(9.13)
3	0.006	(4.65)	3.55	0.162	(.00)	0.0021	(6.22)	.13	0.005	(4.33)
4	0.005	(3.38)	1.58	0.067	(.03)	0.0016	(4.04)	.14	0.005	(3.88)
Total				0.168						
D. Baseline, but without the twenty-seven covariate-tariff interaction instruments										
1	0.015	(7.91)	1.50	0.183	(.20)	0.0041	(7.32)	.19	0.016	(9.71)
2	0.013	(8.72)	2.65	0.282	(.21)	0.0041	(8.29)	.20	0.014	(10.47)
3	0.012	(7.32)	3.42	0.317	(.28)	0.0031	(6.71)	.25	0.012	(9.08)
4	0.011	(2.95)	1.55	0.139	(.49)	0.0011	(2.76)	.20	0.007	(4.02)
5	0.004	(1.15)	1.47	0.042	(.34)	0.0007	(1.97)	.16	0.003	(1.82)
Total				0.150						

Note: This table reports the results of estimating the Angrist-Imbens equations (8) and (9). Column (1) is the IV estimate of the treatment effect (β). Column (3) is the estimated change in exporting induced by the U.S. tariff rate. Column (4) is the effect of exporting on labor productivity for those Canadian plants that were induced to export by the U.S. tariff rate. Column (5) is the overidentification test (χ^2 statistic) and column (6) is the p -value. A p -value below .01 indicates that the exclusion restriction is rejected. Column (7) is the OLS estimate of the treatment effect (β) and column (8) is the t -statistic. "Total" were the employment-weighted average of the $\beta \times \Delta T \times 8$ across bins. The remaining estimated coefficients in equations (8) and (9) appear in Online Appendix Tables B.10-B.12. Column (1') reports the IV coefficient on the treatment T for the case where the bins are redefined using the profit-based method. Sample sizes for Panels A, B, and D are the same as in the notes to Table III. Sample sizes for the four bins in Panel C are between 1,238 and 1,394. The sample consists of 1984 nonexporters.

Read:

- Table V is the preferred heterogeneous-response specification.
- In the baseline panel, the estimated treatment effect β declines across the five bins from 0.012 to 0.002.
- The implied eight-year labor-productivity gains for plants induced to export are 0.147, 0.237, 0.241, 0.085, and 0.022 log points across bins 1 through 5.
- The employment-weighted total effect is 0.107 log points in the preferred baseline.
- Binary-treatment and alternative-bin specifications keep the same monotone pattern.

Economic message. Table V is where the paper's main causal claim lives. The message is not merely that exporting helps; it is that tariff-induced export entry helps mostly the initially smaller and less productive plants.

5.6 Table VI: Canadian input tariffs matter too, but the U.S. market-access channel remains central

Read:

First stage		Second stage		Alternative second stage		Tests	
$\Delta \tau^{Import}$		$\Delta \tau^{Import}, \Delta \ln M$		$\Delta \tau^{Import}, \Delta \ln M$		$T = \ln(EXP_{1996})$	
Coeff	t	Coeff	t	Coeff	t	β	$\Delta \tau$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Baseline (from Table V)							
1	0.012	0.012	0.012	0.012	0.012	0.012	0.012
2	0.010	0.010	0.010	0.010	0.010	0.010	0.010
3	0.009	0.009	0.009	0.009	0.009	0.009	0.009
4	0.005	0.005	0.005	0.005	0.005	0.005	0.005
5	0.002	0.002	0.002	0.002	0.002	0.002	0.002
Total	0.107	0.107	0.107	0.107	0.107	0.107	0.107
B. $\Delta \tau^{Import}$ in both stages							
1	0.008	0.008	0.008	0.008	0.008	0.008	0.008
2	0.010	0.010	0.010	0.010	0.010	0.010	0.010
3	0.005	0.005	0.005	0.005	0.005	0.005	0.005
4	0.004	0.004	0.004	0.004	0.004	0.004	0.004
5	0.002	0.002	0.002	0.002	0.002	0.002	0.002
Total	0.092	0.092	0.092	0.092	0.092	0.092	0.092

- Table VI adds Canadian tariff cuts on U.S. intermediate inputs and related import measures to the first stage and, in some specifications, to the second stage.
- The first-stage coefficients on $\Delta \tau^{Input}$ are positive and often significant, confirming that access to U.S. intermediate inputs also predicts export growth.
- The implied weighted treatment effect rises from 0.107 in the baseline to roughly 0.121–0.128 depending on the specification.

Economic message. The paper’s story is not only “sell more abroad.” Better access to U.S. inputs also supports productivity-enhancing investment. But even after allowing for this extra channel, the main message of heterogeneous export-induced productivity gains survives.

5.7 Table VII and Table VIII: the same plants adopt technology and innovate

TABLE VII TECHNOLOGY ADOPTION AND PRODUCT INNOVATION										
Raw adoption and innovation rates										
	New exporter		Nonexporter		New-Non		OLS		IV	
	(1)	(2)	(3)	(4)	Difference	%	β	t	Difference	t
					(5)	(6)	(7)	(8)	(9)	(10)
					(11)				(12)	
Adoption of advanced manufacturing technologies, 1988–1993										
1. Manufacturing information systems										
Low bin	0.16	0.08	0.10	0.18	0.077	182	0.018	0.018	(3.36)	(3.55)
High bin	0.16	0.17	–0.01	–5	–0.0008	(–0.18)	–0.018	–0.018	(–1.92)	(–0.07)
2. Inspection and communications										
Low bin	0.18	0.10	0.07	72	0.0068	(2.55)	0.021	(3.61)	(3.17)	(0.88)
High bin	0.14	0.20	–0.06	–30	–0.0064	(–1.31)	–0.013	(–1.39)	(–0.05)	(–0.05)
3. Any product or process innovation										
Low bin	0.30	0.20	0.09	46	0.0073	(2.17)	0.021	(2.77)	(2.78)	(0.98)
High bin	0.53	0.57	–0.03	–6	–0.0011	(–0.18)	–0.018	(–1.50)	(–0.06)	(–0.06)
4. Any product innovation										
Low bin	0.26	0.14	0.12	82	0.0083	(2.75)	0.019	(2.77)	(2.31)	(0.67)
High bin	0.43	0.47	–0.05	–10	–0.0022	(–0.34)	–0.011	(–0.90)	(–0.04)	(–0.04)
5. Labor productivity growth, 1988–1996										
Low bin	0.030	0.005	0.024	0.005	(0.82)	0.005	(0.37)	0.005	(0.37)	(0.01)
High bin	–0.005	–0.007	0.002	–0.0001	(–0.16)	–0.002	(–0.81)	–0.002	(–0.81)	(–0.01)

Note: Plants are grouped into four types depending on whether they are (1) new exporters or nonexporters and (2) in low bins (bins 1, 2, and 3) or high bins (bins 4 and 5) of the 1988–1993 period. Column (1) is the low bin, column (2) is the high bin, column (3) is the difference between the low bin and the high bin, column (4) is the percentage difference between the low bin and the high bin, column (5) is the OLS estimate of the treatment effect, column (6) is the OLS estimate of the treatment effect, column (7) is the IV estimate of the treatment effect, column (8) is the IV estimate of the treatment effect. The dependent variable is the binary variable listed in the column heading. Estimates are in parentheses and t-statistics are in parentheses. The first row is the low bin, the second row is the high bin, the third row is the difference between the low bin and the high bin, the fourth row is the percentage difference between the low bin and the high bin, the fifth row is the OLS estimate of the treatment effect, the sixth row is the OLS estimate of the treatment effect, the seventh row is the IV estimate of the treatment effect, the eighth row is the IV estimate of the treatment effect. The sample consists of those 1988 nonexporters that appear in the RAST survey (N = 532). t-statistics are in parentheses.

TABLE VI CONTINUED									
Bin	First stage		Second stage		Alternative second stage		Tests		
	$\Delta \tau^{Import}$		$\Delta \tau^{Import}, \Delta \ln M$		$\Delta \tau^{Import}, \Delta \ln M$		$T = \ln EXP_{1996}$		
	Coeff	t	Coeff	t	Coeff	t	β	$\Delta \tau$	Hausman Overidentification
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
C. $\Delta \tau^{Import}$ and $\Delta \ln M$ in both stages									
1	43.3	(7.60)	-11.54	(-2.87)	0.007	(3.25)	0.02	(1.48)	0.127
2	27.2	(4.33)	-3.30	(-0.68)	0.009	(4.66)	-0.02	(-0.17)	0.127
3	37.2	(6.25)	-2.17	(-0.45)	0.008	(3.39)	0.07	(0.82)	0.127
4	23.9	(2.84)	-0.88	(-0.11)	0.004	(1.92)	0.15	(1.38)	0.127
5	-0.4	(-0.05)	3.80	(0.78)	0.001	(0.79)	0.02	(0.21)	0.127
Total									0.127
D. $\Delta \tau^{Import}$ and $\Delta \ln M$ in both stages									
1	39.7	(7.14)	-3.97	(-1.71)	0.008	(3.35)	0.29	(1.97)	0.148
2	26.5	(4.31)	-2.35	(-0.76)	0.010	(3.66)	0.00	(-0.01)	0.145
3	36.7	(6.36)	-1.15	(-0.42)	0.007	(3.79)	0.13	(1.07)	0.147
4	22.7	(2.82)	0.47	(0.12)	0.004	(1.93)	0.15	(1.37)	0.146
5	1.3	(0.17)	-2.64	(-0.75)	0.002	(0.95)	0.05	(0.70)	0.149
Total									0.121

Note: Each row of this table reports estimates of the second-stage equation (8) and the first-stage equation (9). The second-stage dependent variable is labor productivity growth $\Delta \tau$. The first-stage dependent variable is the treatment $T = 0$. In EXPT₁₉₉₆. Panel A is the baseline specification carried over from Table V. Panels B–D augment this specification with additional regressors. This is most clearly explained with reference to Panel C. B introduces two more regressors into the first stage. These are the plant-specific Canadian tariff rate both on intermediate inputs purchased by the plant ($\Delta \tau^{Import}$) and on commodities produced by the plant ($\Delta \tau^{Export}$). Columns (1) and (3) report the estimated first-stage coefficients on $\Delta \tau^{Import}$ and $\Delta \tau^{Export}$, respectively. Columns (5)–(10) report second-stage coefficients when $\Delta \tau^{Import}$ and $\Delta \tau^{Export}$ are included in the second stage. Columns (11)–(14) report second-stage coefficients when $\Delta \tau^{Import}$ and $\Delta \tau^{Export}$ are omitted from the second stage. Columns (15) and (16) report Hausman and Breusch-Lagan overidentification statistics, respectively, for the alternative second-stage specifications (columns (11)–(14)). The Breusch-Lagan statistics are tiny, indicating that the Canadian tariff rate are exogenous. Turning to differences across panels, Panel B omits $\Delta \tau^{Export}$, but keeps $\Delta \tau^{Import}$. Panel D replaces $\Delta \tau^{Export}$ with $\Delta \ln M$, the growth in Canadian imports from the United States on commodities produced by the plant. The sample consists of 1988 nonexporters. t-statistics are in parentheses.

Adoption of advanced manufacturing technologies, 1988–1993		Engagement in innovative activities, 1988–1991		Any product innovation		Labor productivity growth, 1988–1996	
Manufacturing information systems (MIS)	Inspection and communication	Any innovation	Any product innovation	Any product innovation	Any product innovation	Any product innovation	Any product innovation
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Baseline (from Table VII)							
0.018	(3.36)	(3.55)	0.021	(3.61)	(3.17)	0.021	(2.77)
B. $\Delta \tau^{Import}$ as an instrument							
0.018	(3.49)	(3.21)	0.018	(3.13)	(2.59)	0.019	(2.73)
C. $\Delta \tau^{Import}$ and $\Delta \ln M$ as instruments							
0.019	(3.78)	(3.64)	0.017	(3.10)	(3.14)	0.018	(2.63)
D. $\Delta \tau^{Import}$ and $\Delta \ln M$ as instruments							
0.017	(3.45)	(3.15)	0.017	(3.14)	(3.12)	0.019	(2.73)

Note: This table reports IV estimates of $\Delta \tau$, the low bin coefficient on the treatment T in the second-stage equation (8). The dependent variable is the binary variable listed in the column heading. Estimates are in parentheses and t-statistics are in parentheses. The first row is the low bin, the second row is the high bin, the third row is the difference between the low bin and the high bin, the fourth row is the percentage difference between the low bin and the high bin, the fifth row is the OLS estimate of the treatment effect, the sixth row is the OLS estimate of the treatment effect, the seventh row is the IV estimate of the treatment effect, the eighth row is the IV estimate of the treatment effect. The sample consists of those 1988 nonexporters that appear in the RAST survey (N = 532). t-statistics are in parentheses.

Read:

- In the low bin, new exporters adopt MIS at a rate of 16% versus 6% for nonexporters, a difference of 183%.
- In the low bin, new exporters adopt inspection and communications technologies at a rate of 18% versus 10%.
- In the low bin, any innovation is 30% versus 20%, and product innovation is 26% versus 14%, so product innovation is 82% more likely among new exporters.

TABLE IX
EVIDENCE THAT LABOR-PRODUCTIVITY GAINS REFLECT TFP GAINS

Bin	β (1)	t (2)	$\beta \times \Delta T \times 8$ (3)	N (4)	Bin	β (5)	t (6)	$\beta \times \Delta T \times 8$ (7)	N (8)
A. Domestic (Canadian) sales					B. Inventories of intermediate inputs				
1	0.0064	(3.18)	0.08	979	1	-0.0022	(-7.80)	-0.03	979
2	0.0060	(4.30)	0.14	953	2	-0.0010	(-4.44)	-0.02	853
3	0.0031	(2.32)	0.09	1,208	3	-0.0010	(-5.62)	-0.03	1,208
4	0.0029	(1.72)	0.05	1,020	4	-0.0007	(-3.64)	-0.01	1,020
5	0.0022	(1.00)	0.03	1,072	5	-0.0002	(-1.07)	0.00	1,072
Total			0.057		Total			-0.013	
C. Intermediate input purchases/shipments					D. Energy purchases/shipments				
1	-0.0074	(-4.45)	-0.09	980	1	-0.0086	(-4.73)	-0.11	980
2	-0.0049	(-4.56)	-0.11	951	2	-0.0109	(-7.94)	-0.20	951
3	-0.0044	(-3.97)	-0.12	1,205	3	-0.0092	(-6.64)	-0.26	1,205
4	-0.0034	(-1.77)	-0.05	1,020	4	-0.0056	(-2.63)	-0.09	1,020
5	0.0002	(0.14)	0.00	1,071	5	0.0025	(1.27)	0.03	1,071
Total			-0.052		Total			-0.092	

Note: This table reports IV estimates of equation (5), except that the dependent variable is now the 1985-1995 average annual log change in the variable indicated in the heading of the panel, such as, the 1985-1995 average annual log change in domestic (Canadian) sales, β in column (1) and (5) is the coefficient on the treatment variable ($T = 0$ is EXPG). Aside from the dependent variable, the specification is identical to that in the baseline specification of Table V. The predicted treatment used for IV and for the change in treatment ΔT used in columns (3) and (7) comes from equation (6) and uses the same specification as in the baseline specification of Table V. The sample consists of 384 heterogeneous exporters are top producers.

- The corresponding IV estimates are positive and significant in the low bin, but close to zero in the high bin.
- Table VIII shows that these conclusions are robust when Canadian tariff cuts and import-growth variables are added.

Economic message. This is the paper's key mechanism section. The firms that gain in productivity are exactly the firms that invest in productivity-enhancing technologies and innovation. That makes it much harder to tell a pure mark-up story.

5.8 Table IX: why the labor-productivity result likely reflects real efficiency gains

Read:

- Panel A shows that the same induced exporters increase domestic Canadian sales, especially in the low bins.
- Panel B shows that inventories of intermediate inputs fall, consistent with better inventory management and MIS adoption.
- Panels C and D show declines in intermediate-input purchases per shipment and energy purchases per shipment, again especially for the lower bins.

Economic message. If the labor-productivity result were only a mark-up effect, new exporters should lose domestic sales. Instead they gain sales and reduce inputs per unit of output. This is some of the most important indirect evidence in the paper that true efficiency improved.

5.9 Table X and Table XI: old exporters benefit too, but without negative selection

Read:

- Table X shows that among old exporters, firms with higher export growth have higher labor-productivity growth, and the OLS treatment coefficient is about 0.060 in all three bins.

TABLE X
LABOR PRODUCTIVITY GROWTH 1988–1996, OLD EXPORTERS SAMPLE: OLS

	Labor productivity growth differences		OLS			
	Mean	<i>t</i>	β	<i>t</i>	R^2	<i>N</i>
	(1)	(2)	(3)	(4)	(5)	(6)
All plants	0.018	(4.30)	0.060	(4.04)	.15	1,607
1. Productivity and employment below median	0.021	(2.22)	0.061	(1.70)	.11	351
2. Other	0.011	(1.87)	0.060	(2.84)	.15	802
3. Productivity and employment above median	0.023	(3.38)	0.059	(2.23)	.09	454

Notes. This table reports results for the sample of old exporters, plants that exported both in 1984 and in 1996. The treatment is defined as the average annual log change in exports, $T = \ln(\text{EXP}_{1996}/\text{EXP}_{1984})/12$. Columns (1) and (2): Divide plants into two groups based on whether the plant has a T that is above or below 0.075 (the median of T). Column (1) reports the *difference* between the two groups in labor productivity growth Δy (above median minus below median). Columns (3)–(5): These report OLS estimates of equation (8). The coefficient on T is reported in column (3). The specification of equation (8) differs from that reported in our baseline specification (Panel A of Table V) in three ways: (i) the sample consists of old exporters, (ii) the treatment is $T = \ln(\text{EXP}_{1996}/\text{EXP}_{1984})/12$, and (iii) industry fixed effects are defined at the two-digit SIC level.

TABLE XI LABOR PRODUCTIVITY GROWTH 1988–1996, OLD EXPORTERS SAMPLE: IV									
Second stage					First stage			Tests	
Treatment T					Δy^{import}		$\Delta y^{\text{import}}, \Delta \ln M$		Hausman (9)
β	<i>t</i>	ΔT	$\beta \times \Delta T \times 8$		Coeff	<i>t</i>	Coeff	<i>t</i>	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(10)	(10)
A. Baseline (Δy^{import} , Δy^{import} , and $\Delta \ln M$ omitted from both stages)									
0.33	(3.46)	0.026	0.067						6.80
B. Δy^{import} as an instrument									
0.27	(3.45)	0.040	0.088	0.53	(4.80)				6.64
C. Δy^{import} and Δy^{import} as instruments									
0.23	(2.92)	0.042	0.075	0.46	(3.42)	0.17	(1.80)	4.41	1.00
D. Δy^{import} and $\Delta \ln M$ as instruments									
0.29	(3.74)	0.044	0.102	0.49	(3.75)	0.06	(1.34)	7.90	0.78

Notes. This table is similar in structure to Table VI. The alternative second stage in Table VI corresponds to the second stage in this table. The table reports estimates of equations (6) and (9). The dependent variable is 1988–1996 labor productivity growth (Δy). The treatment is $T = \ln(\text{EXP}_{1996}/\text{EXP}_{1984})/12$, the average annual log change in exports. The overidentification χ^2 statistic in column (10) are all small, indicating that U.S. and Canadian tariff rates are exogenous. See the notes to Table VI for an explanation of the remaining column and rows. The sample consists of the set of plants that exported in both 1984 and 1996 ($N = 1,607$). *t*-statistics appear in parentheses.

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- The coefficient equality across bins is essentially exact: there is no negative selection for old exporters.
- Table XI reports IV estimates for old exporters. In the baseline, $\beta = 0.33$ and the implied eight-year treatment effect is 0.067 log points.
- When Canadian input tariffs are added as instruments, the implied effect rises to about 0.088 log points.

Economic message. These tables are an elegant confirmation of the model. New exporters show declining treatment effects with initial productivity; old exporters do not. That pattern is hard to rationalize with a generic omitted-variable story and fits the theory unusually well.

6 Short summary

- The paper studies how the Canada-U.S. FTA changed Canadian plant productivity through improved foreign market access.
- It builds a model where exporting and investing in productivity are complements and where treatment effects differ across plants.
- Using plant-specific tariff cuts as instruments, it shows that plants induced to export have higher labor productivity.
- These gains are concentrated among initially smaller and less productive nonexporters, exactly as predicted by the model’s negative-selection mechanism.
- The same plants also adopt advanced manufacturing technologies and engage in product innovation.
- Additional evidence from domestic sales, inventories, and input usage suggests the labor-productivity gains reflect real efficiency improvements rather than just higher mark-ups.
- Old exporters also gain from improved market access, but their treatment effect is roughly homogeneous across initial productivity bins.
- The paper concludes that the FTA raised Canadian manufacturing productivity through both reallocation across plants and within-plant productivity improvements.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that better access to foreign markets can raise plant-level productivity, but only for some plants. The gains are not spread evenly across all firms. Rather, the main beneficiaries are the plants for which the FTA made exporting and productivity investment jointly worthwhile. This is shown with a rare combination of theory, plant-level administrative data, and an instrument that is constructed at the plant-product level.

7.2 Economic intuition

The economic intuition is simple but powerful. Exporting expands the market over which a productivity improvement can be amortized. For plants that are close to the relevant margin, better access to the U.S. market flips the export-and-invest decision from unprofitable to profitable. These are exactly the plants that start exporting, adopt better production technologies, innovate more, and become more productive. The initially strongest firms are different: many were already exporting, so the joint margin is gone and the treatment effect is more homogeneous.

7.3 Takeaway

The deepest takeaway is that *market access is an innovation incentive*. Trade policy matters not only because it reallocates sales across firms, but also because it changes which firms find it worthwhile to invest in becoming better producers. That is why the paper remains so important: it shows that trade liberalization can generate within-firm productivity gains, and it shows very clearly which firms those gains belong to.